

Tomato



TOMATO (*Lycopersicon esculentum)*

Climate and soil

The crop is grown from almost MSL to an altitude of 1500 m in tropical and subtropical regions, with an annual rainfall of 60-150 cm. Very high rainfall during its growth is harmful. When grown under hot weather, it is cultivated as an irrigated crop. The winter crop is planted from August to September. For organic farming of tomato winter crop has been found to be ideal.
Well drained sandy loam soil with high level of organic contents is best suitable for tomato cultivation. Soils with high acidity are not suitable for tomato cultivation. Three to 4 q of suitable lime can be applied in the field in an interval of three years to reduce the level of acidity to tolerable limits. There is a need to go for soil testing at the beginning of the crop season.

Land preparation

The land may be ploughed and harrowed 3 or 4 times to obtain a fine tilth. About 10 tonnes of Farm Yard Manure (FYM) or vermi compost/compost @ 1-1.5 t per acre is applied at the last ploughing. Green manuring is recommended for areas with assured rainfall and also for irrigated crop. Beds are prepared against the slope and after levelling the bed, field channels of 50 cm breadth are prepared at intervals of 1 m.

Varieties

Sioux

- It is a high yielding American variety with dwarf, spreading and determinate growth habit.
- The fruits are medium to large with whitish green colour when unripe, and uniform red colour on ripening.
- It yields about 250 q/ha.

Pusa Red Plum

- It has been evolved at the IARI by a cross between a cultivated tomato and the wild *L .pimpinellifolium.*
- It is an early variety suitable for table use.
- The fruits are small, borne in clusters with red colour and high vitamin C and sugar content.
- The plants are tall, with determinate growth habit, vigorous and high yielding.

Pusa Early Dwarf

- It is an early ripening selection from the cross between 'Improved Meeruti' and 'Red Cloud' - a typical dwarf type with medium large fruits of uniform colour.
- It has been observed to do well in both the seasons, and yields up to 395 q/ha.

Pusa Ruby

- It is an early and hardy variety evolved at IARI. The plants are medium and determinate.
- The fruits are medium-sized, lobed, and uniformly red when ripe and yield 330 q/ha around Bangalore.
- Some other varieties like 'Sel-120' and 'Sel-152' have been recently recommended by IARI.

Co-1

- The new variety was recommended by the former Department of Agriculture, Tamil Nadu, for the state.
- The plant is dwarf having semi-spreading habit with dark-green foliage.
- The fruits are round and smooth, pale green when unripe and capsicum red when fully ripe.
- The fruits are borne in cluster of 6 to 8, each weighing about 125 g. It yields an average of 380 q/ha.

CO 2

- It is a selection from a Russian introduction. It is an indeterminate variety.
- The fruits are flat with 4-5 furrows.
- Each fruit weighs on an average 55-60 g.
- Ripe fruits are deep orange red in colour, its duration is 140 days and the yield is 28-30 tonnes/ha.

CO 3 (Marutham)

- It is an induced mutant from Co 1; a determinate type suitable for close planting even as close as 30 cm x 30 cm.
- Fruits are round in shape with four light grooves and capsicum red in colour.
- It is a cluster bearing type capable of yielding as high as 40 tonnes/ha under a close spacing in a duration of 100-105 days.

S-12

- The Punjab Agricultural University recommended the new round-fruited variety as a high-yielding variety for the State of Punjab.
- It has been developed through irradiation.

Punjab Chhuhara

- Medium-Size, pear-shaped fruit and a high-yielding variety released by the Punjab Agricultural University.
- 'Sweet-72'. Developed at the Regional Agricultural Research Institute, Gwalior.
- It is a cross between 'Rusa red plum' and 'Sioux'.
- It bears round, scarlet red, sweet fruits.
- Yield is about 200 q/ha.

PKM 1

- This is another induced mutant from a local variety called Annanji.
- The fruits are flat-round with prominent green shoulder and capsicum red colour.
- It is suitable for long distance transport.
- It yields about 30-35 tonnes/ha in a crop duration of 135 days.

Pusa Ruby

- It is an indeterminate variety suitable for irrigated and also rainfed cultivation.
- The fruits are flat, medium in size with 7-8 furrows and capsicum red in colour.
- It yields about 25-30 tonnes/ha.

Paiyur-1

- This variety was evolved by crossing Pusa Ruby and CO 3.
- It is also suitable for rainfed cultivation.
- It yields about 30 tonnes/ha.

Arka Saurabh

- Developed by Indian Institute of Horticultural Research, Bangalore, it is a high-yielding variety exhibiting 30-35% higher yield than "Pusa Ruby'.
- It is semi-determinate, bears very firm, deep red, medium-sized fruits with smooth skin and thick flesh.
- The fruits are round with nipple tip.
- It has very good transport and keeping quality, 8-10 days at room temperature amongst the round-fruited varieties.
- It has an excellent blend of sugar and acidity and is very suitably both for fresh and processing purposes particularly for ketchup and juice-making.

Arka Vikas

- Another high-yielding variety with large-sized, oblate, medium firm fruits.
- It is suitable for fresh market.

Arka Ahuti (Sel.11)

- It is pureline selection from 'Ottawa-60' developed at IIHR, Bangalore.
- Plants are semi-determinate with a duration of 135 days.
- The fruits are oblong, thick fleshed, bilocular and uniform ripening.
- The fruits have a TSS of 5.25% with an acidity of 0.35%.
- It has an yield potential of 45 tonnes/ha.

Arka Ashish (IHR 674 SBSB)

- Developed at IIHR, Bangalore.
- It is an improvement over UC83 B from California.
- The plants are determinate with concentrated fruit maturity.
- The fruits are uniform ripening, very firm, bilocular, oval and the ripened fruits store well on the vine.
- It is tolerant to powdery mildew.
- The colour is excellent (Lycopene 10 mg/100g), TSS is 4.8% and with low acidity (0.35%), it makes excellent ketchup, juice and puree.
- It yields 35 tonnes/ha. in 130 days.

Arka Abha (BWR-1)

- This is from line selection (IHR 638-12) from a SSD derived line got from AVRDC, Taiwan (VC8-1-2-1).

- The semi determinate vines bear oblate fruits with an average fruit weight of 75 g with light green shoulder.
- It is resistant to bacterial wilt and is capable of yielding 25 tonnes/ha. in 135 days.

Arka Alok (BWR-5)

- It is also a bacterial wilt resistant variety developed through pure line selection (IHR 719-1/6) out of an accession (CL114-5-1-0) from AVRDC, Taiwan developed at IIHR, Bangalore.
- The plant habit is determinate with a duration of 125 days.
- The fruit weighs on an average of 100 g and is square round in shape, thick fleshed with light green shoulder.
- It has an yield potential of 35 tonnes/ha.

Shakthi

- This was developed at College of Horticulture, Kerala Agricultural University, Vellanikkara.
- It is resistant to bacterial wilt disease. It has an yield potential 0f 30 tonnes/ha.

HS101

- Developed at Haryana Agricultural University, Hisar by hybridization (Sel 2-3 X Exotic culture) and selecting in the segregating progenies.
- The plants are determinate, dwarf, fruit medium sized, round, borne in cluster of 2-3, juicy, suitable for winter season for northern region.

HS102

- Developed at HAU, Hisar by selecting from the segregating progenies of the cross S12 X Pusa Early Dwarf.
- It is an early variety with concentrated flowering and fruiting.
- Fruits medium to small, round, juicy, thin pericarp, recommended for cultivation in Haryana.

HS110

- A late variety developed at HAU, Hisar through selection from an exotic line; plants are determinate, potato leaved, fruits are large, smooth, fleshy with jointless pedicel suitable for table purpose. Recommended for cultivation in Haryana.

Hisar Arun (Sel-7)

- Developed at HAU, Hisar, by hybridization and selection from a cross Pusa Early Dwarf X K1.
- Extremely early and high yielding, plants are determinate, dwarf in stature with concentrated flowering and fruiting; fruits medium to large, round, deep red, recommended for cultivation in Delhi, Uttar Pradesh, Karnataka, Orissa and Maharashtra.

Hisar Lalima (Sel-18)

- Developed from the segregating population of a cross Pusa Early Dwarf X HS101 at HAU, Hisar.
- An early variety. Plants are determinate, leaves have deep cut, fruits are round, large, fleshy, high yielding.

Hisar Lalit (NRT 8)

- Developed at HAU, Hisar by bybridization and selection from a cross HS101 X Resistant Bangalore.
- The plants are semi-determinate, early with round medium to large fruits, resistant to root-knot nematode.
- Recommended for cultivation in root-knot nematode infested area.

Hisar Anmol (H24)

- It is a variety showing field resistance to tomato leaf curl virus. Developed at HAU, Hisar through selection from the cross Hisar Arun X *L.hirsutum f.glabratum.*
- Plants are determinate, fruits medium in size, round, red and fleshy.

KS.2

- An early variety developed at Kalyanpur, Kanpur; plants determinate, fruits flattis round, slightly furrowed with green stem end.

SL 120

- It was developed at IARI, New Delhi.
- Plants are semi-determinate, fruits are large, round, fleshy, less acidic and less seeded.
- This variety is resistant to root-knot nematode.

Pusa Gaurav

- It was developed at IARI, New Delhi. The plants have cut leaves with light green colour.
- Fruits are yellowish red, smooth, oblong, bilocular, uniform ripening, good for processing and long distant transport.

S 12

- Developed at Punjab Agricultural University, Ludhiana. Plants are dwarf bushy, vigorous; fruits medium sized, compressed, round, uniform red at maturity, juice and acidic.

Pant Bahar (AC 238)

- Developed at Govinda Ballaba Pant University of Agriculture and Technology, Pant Nagar.
- Plants are bushy, branched, light green foliage, small leaflets, flattish round medium sized fruits, slightly ridged.

Pant T3

- Another variety developed through pure line selection at G.B.Pant University, Pant Nagar.
- Plants are semi-determinate, stem thick, round, hairy.
- Suitable for winter season, fruits medium, round and smooth.
- Recommended for Delhi, Uttar Pradesh, Orissa and Maharashtra.

Narendra Tomato 1 (NDT 5)

- Developed at Narendra Dev Agricultural University, Narendra Nagar, Faizabad through a selection from Exotic genotype (EC 130041).
- Plants are indeterminate, fruits large in size (85-110 g/fruit), round, smooth with uniform red colour at maturity, less juice, fleshy, suitable for table purpose and processing.
- Released by the Uttar Pradesh state variety release committee in 1996.

Narendra Tomato 2 (NDT 120)

- Another variety developed at Narendra Dev Agricultural University from an exoctic line EC 119201.
- Plants are determinate with medium sized round fruits having a slightly pointed stigmatic end, pericarp thick.
- Average fruit weight 55-75 g, good for transport and processing.

Solan Gola

- Developed at Y.S.Parmar University of Horticulture and Forestry, Solan, Himachal Pradesh.

Arka Meghali

- It is a new high yielding tomato variety developed at IIHR, Bangalore through pedigree method of selection from a cross between Arka Vikas and IHR 554 specifically suited for rainfed cultivation.
- Under rainfed situation it recorded on an average 17.2 tonnes/ha compared to 10.9 tonnes/ha by Pusa Ruby.
- The individual fruit weight is 63 g.

PKM 1, CO 3 (Marutham) and Paiyur 1, Arka Abhijit, Arka Abha, Arka Ananya, Arka Alok, Arka Shreshta, Arka vardan, Arka Vishal, Arka Vikas, Arka Saurab, Arka Meghali and Arka Ahuti are some of the popular varieties.

Season of sowing

Planting is done during May - June and November – December

Seed rate

About 400g of seeds is required for a hectare.

Seed treatment

Treat the seeds with *Trichoderma viride* 4 g or *Pseudomonas fluorescens* 10 g or Carbendazim 2 g per kg of seeds 24 hours before sowing. Just before sowing, treat the seeds with 40 g *of Azospirillum*.

Nursery preparation

Apply FYM 10 kg, Neem cake 1 kg, VAM 50 g, Super phosphate 100 g and Furadon 10 g per square metre during the nursery area preparation. Sow the seeds in lines at 10 cm apart in raised seed nursery beds and cover with sand. Irrigation is done using rose can and the seedlings are transplanted 25 days after sowing.

Preparation of field

Plough the land to a fine tilth. Apply 2 kg/ha of Azospirillum and Phosphobacterium mixed with 100 kg FYM before planting and form ridges and furrows. Form ridges and furrows and transplant 25 days old seedlings on one side of the ridges.

Application of fertilizer

Apply FYM 25 t/ha, N 75 kg, P 100 kg, K 50 kg, Borax 10 kg and Zinc sulphate 50 kg/ha as basal dose and 75 kg N/ha on 30th day of planting during earthing up. Spray 1.25 ppm (1.25 mg in one lit) Triacontanol, 30 days after transplanting and at full bloom stage to increase the yield.

Weed control

Apply Pendimethalin 1.0 kg a.i./ha or Fluchloralin 1 kg a.i./ha as pre-emergent herbicide followed by one hand weeding on 30 days after planting. Mulching with black Low Density Polyethylene (LDPE) sheets of 25 micron thickness by burying both the ends into the soil to a depth of 10 cm will avoid weed growth.

Yield

PKM 1 : 30-35 t/ha
CO 3 : 40 t/ha
Paiyur : 30 t/ha

HYBRID TOMATO
Varieties
COTH 1 and COTH 2

Season
Planting is done during May – June and November – December
Seed Rate
About 150 g of seeds are required for a hectare
Seed Treatment
Treat the seeds with *Trichoderma viride* @ 4g/kg of seeds 24 hours before sowing and with *Azospirillum* @ 20 g/150 g just before sowing.
Nursery preparation
The seedlings are raised under 50% shade net nursery unit. Seeds can be sown in 0.8 mm thick protrays (98 cells) filled with 1.25 kg composted coir pith. The seeds are sown @ one seed per cell and kept in the raised beds. The seedlings are transplanted 25 days after sowing.
Field preparation and planting
Prepare the field with the addition of FYM @ 25 t/ha and form ridges and furrows at a spacing of 60 cm. Apply each 2 kg/ha (10 packets) of *Azospirillum* and *Phosphobacteria* by mixing with 20 kg of FYM. Irrigate the furrows and transplant the seedlings, with the ball of earth, on the ridges adopting a spacing of 45 cm between the plants.
Manuring
Basal Dose : NPK - 50:300:50 kg/ha
Top dressing: N and K each 150 kg/ha in 3 equal splits at 30, 45 and 60 days after planting.
Irrigation
Irrigate the field on the third day of planting and subsequently at weekly intervals.
Weed control
Apply Pendimethalin 1.0 kg a.i./ha or Fluchloralin 1 kg a.i./ha as pre-emergence followed by one hand weeding on 30 days after planting. Mulching with black Low Density Polyethylene (LDPE) sheets of 25 micron thickness by burying both the ends into the soil to a depth of 10 cm will avoid weed growth.
After cultivation

- Stake the plants 30 days after planting with 1 - 1.5 m tall stakes.
- Remove the side branches up to 20 cm from ground level.
- Foliar spray of ZnSO4 (0.5 per cent) thrice at 10 days interval from 40 days after planting.

Harvest: Harvest the fruits when they turn red.
Duration: 110 - 115 days from transplanting (135 - 140 days from sowing).
Yield : Average yield is 80-90 t/ha
PLANT PROTECTION
PESTS
Fruit borer *(Helicoverpa armigera* and *Spodoptera litura*)
Management:

- Grow simultaneously 40 days old American tall marigold and 25 days old tomato seedlings @ 1:16 rows.
- Set up pheromone traps @ 12/ha.
- Collection and destruction of damaged fruits and grown up caterpillars.
- Spray Endosulfan 35 EC 2 ml/lit or Carbaryl 50 WP 2 g/lit or *Bacillus thuringiensis* 2g/lit or Quinalphos 2.5 ml/lit.
- Release *Trichogramma chilonis* @ 50000/ha/release coinciding with flowering time and based on ETL.

For *Helicoverpa armigera*: Helicoverpa armigeraNuclear Polyhedrosis Virus (H.a.NPV) 1.5 x 1012 POBs/ha
For *Spodoptera litura*: Spodoptera litura Nuclear Polyhedrosis Virus (S.l. NPV)1.5 x 1012 POBs/ha5.
Provide poison bait with Carbaryl 1.25 kg, rice bran 12.5 kg, jaggery 1.25 kg and water 7.5 lit/ha.
Serpentine leaf miner
Serpentine leaf miner can be controlled by spraying Neem Seed Kernel Extract 50 g/lit.
Whitefly

- Install yellow sticky traps to attract the adult.
- Spray Dichlorvos 76 WSC @ 1 ml/lit or Triazophos 40 EC 2 ml/lit or fish oil rosin soap 25 g/lit or Dimethoate 2 ml/lit or Methyl demeton 25 EC 2 ml/lit along with the wetting agent (Teepol@ 1 ml/lit)
- Remove alternate weed host *Abutilon indicum*

Nematode
Application of Carbofuran 3 G at 10 g/sq.m at sowing and 1 kg a.i./ha in the main field one week after transplanting. Treat the seeds with antagonistic fungi *Trichoderma viride* or *T. harzianum* at 4 g/ha seed along with pressmud at 5 kg/m2 or Carbofuran 3G at 10 g/m2 to control rootknot nematode.
DISEASES
Damping off (nursery)
Treat the seeds with *Trichoderma viride* 4 g/kg or *Pseudomonas fluorescens* 10 g /kg of seed 24 hours before sowing. Apply *Pseudomonas fluorescens* as soil application @ 2.5 kg/ha mixed with 50 kg of FYM. Water stagnation should be avoided. Drench with Copper oxychloride at 2.5 g/lit at 4 lit/sq.m.
Leaf spot
Leaf spot can be controlled by spraying Zineb or Mancozeb 2 g/lit.
Leaf curl
Spray systemic insecticides like Methyl demeton or Monocrotophos or Dimethoate at 2 ml/lit. to kill the insect vector (whitefly).

Tomato spotted wilt disease
Apply Carbofuran 3 G 1 kg. a.i./ha in nursery at sowing and second application at 1.25 kg a.i./ha 10 days after transplanting in mainfield and 3 sprays of Endosulphan 35 EC 1.5 ml/lit @ 25, 40, 55 days after transplanting.

Agro Advisory Service



Focus Districts

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